



Overview of Development Platforms

Dr. Jim D. Pham

Ohlone College

jpham@ohlone.edu

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Topics

- Application Development Platforms
- Competing Application Development Platforms
- Overview of Application Architectures
- Popular Programming Languages: C++, Java, and .NET
- Overview of Integrated Development Environments
- 15 Minutes Application using The Microsoft .NET
- Q & A



Application Development Platforms

- Android application
- iPhone and iPad application
- Linux application
- Web application
- Windows application
- Windows mobile application



Competing Application Development Platforms

- The Microsoft .NET Development Platform
 - Visual Studio 2010
- The Java EE Platform
 - Oracle JDeveloper, Eclipse, etc.
- C/C++
 - Windows Visual C++ Platform,
 - the Eclipse C/C++ Development Toolkit
- Cocoa with the Xcode IDE
 - Mac, iPhone, and iPad applications
- Android
 - Mac, iPhone, and iPad applications



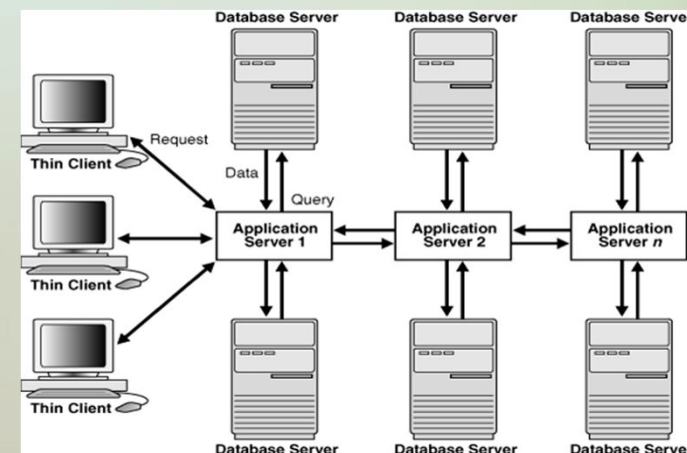
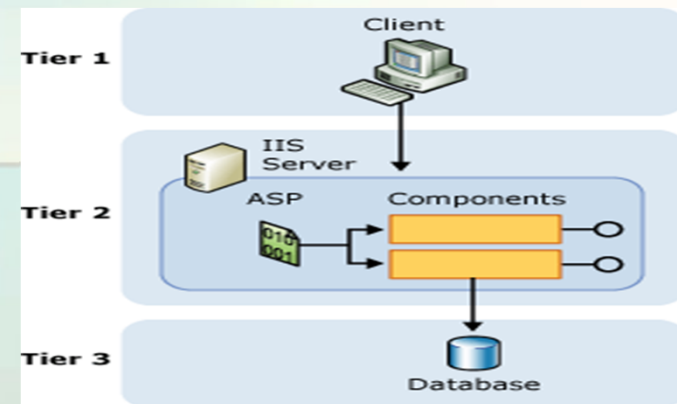
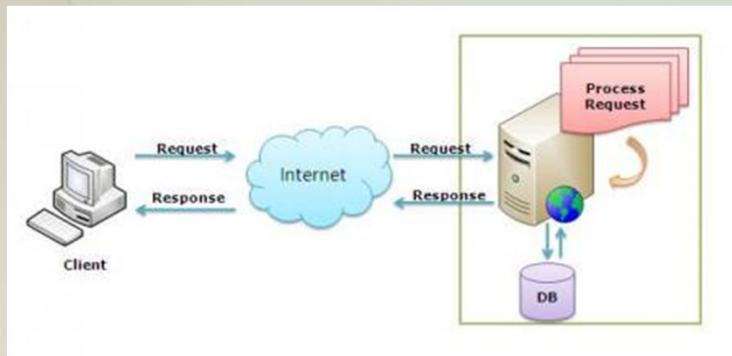
Application Architectures

- Standalone architecture
- Client-Server (or 2-tier) architecture
- Multi-tier architecture
- Service Oriented Architecture (SOA)
- Virtualization and cloud computing



Multi-tier/Distributed Application Architecture

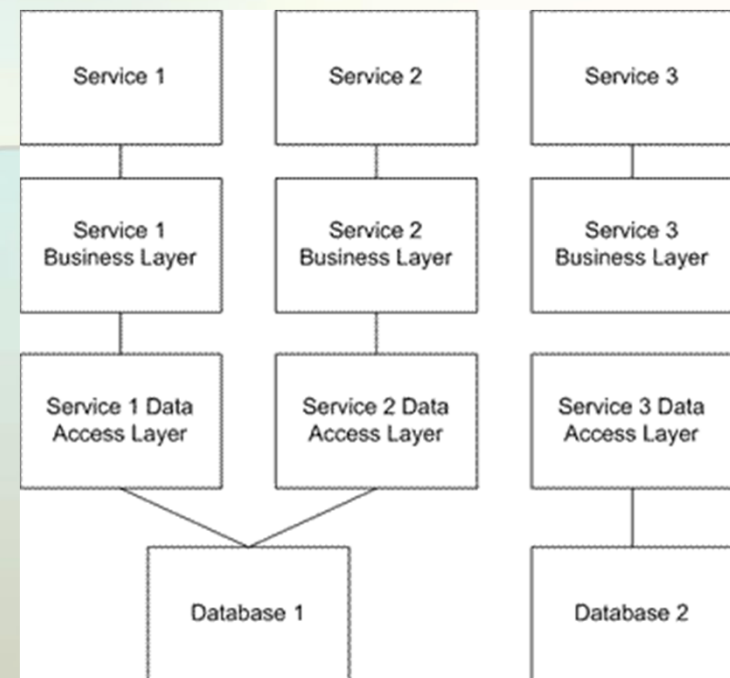
- Client/server applications
 - Server sends the Web Pages/Windows to the Client.
 - Client displays the Web Pages/Windows in Browser.





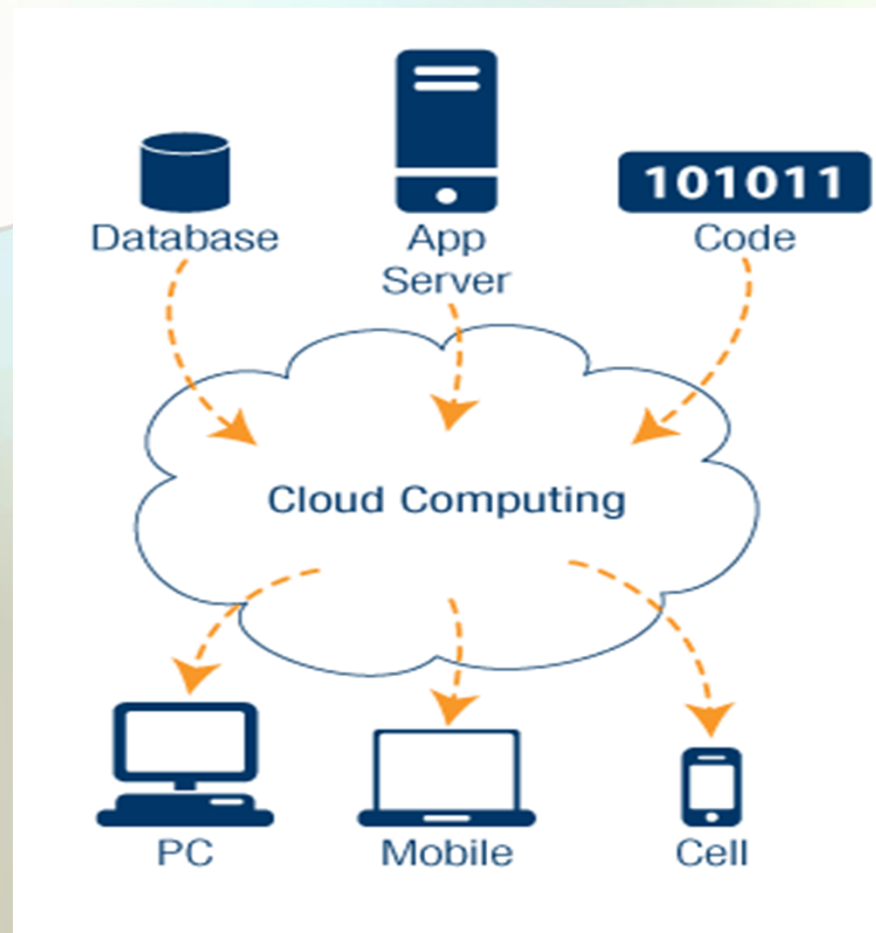
Service Oriented Architecture (SOA)

- Flexibility - enable developers to quickly build new applications
- Reusability - employ investments across multiple applications
- Loose coupling - eliminate and avoid the brittle nature of monolithic applications





Cloud Computing





Popular Programming Languages Comparison

	C++	Java	.NET (C#, VB, etc.)
Encapsulation / Information Hiding	Yes	Yes	Yes
Inheritance	Yes	Yes	Yes
Polymorphism	Yes	Yes	Yes
All user-defined types are Objects	No	Yes	Yes



Class Definition Comparison

Using .NET: C#

```
namespace com.ohlone.edu.demo
{
    class Student
    {
        public string firstName;
        public string lastName;
        // ...
    }
}
```

Using Java

```
package com.ohlone.edu.demo;

public class Student
{
    public string firstName;
    public string lastName;
    // ...
}
```

How is C++ differ from C# and Java?

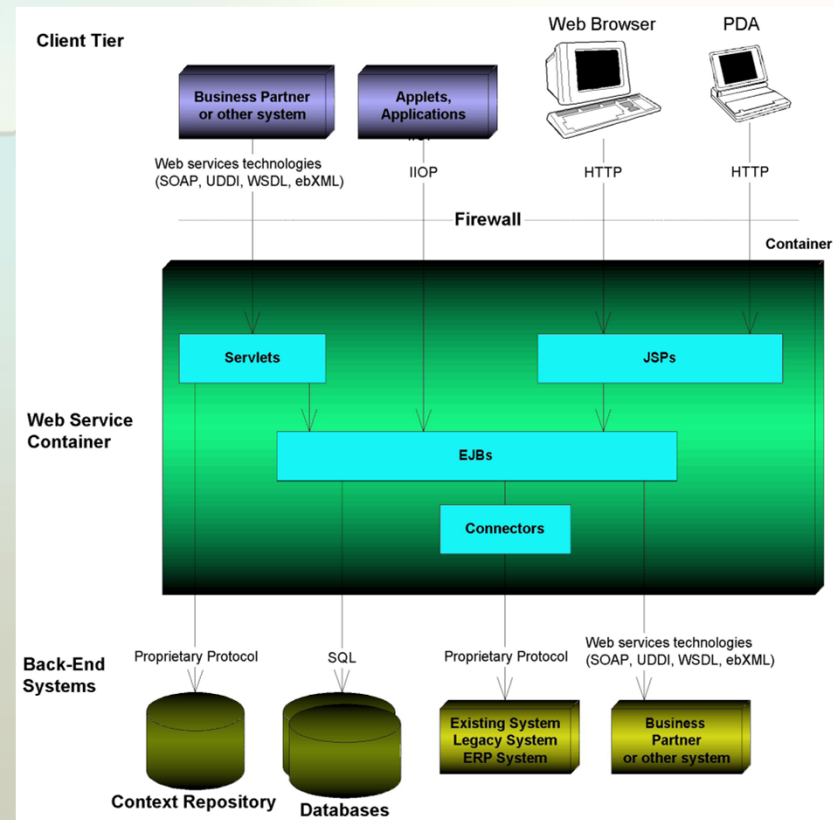
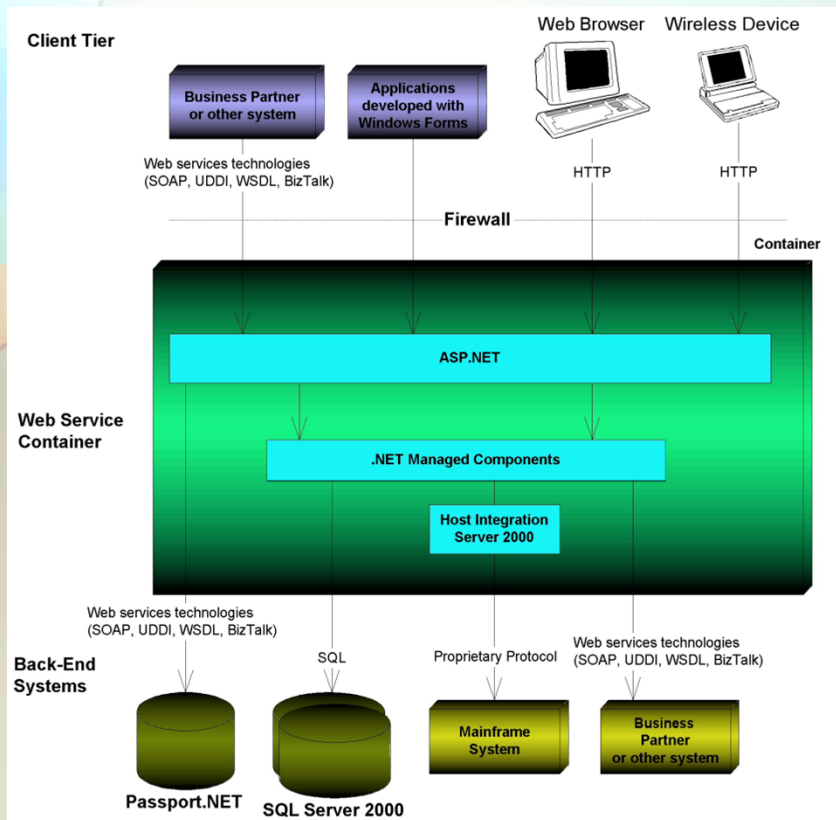


Java EE and .NET technologies Comparison

Feature	Java EE	.NET
Type of technology	Standard	Product
Middleware Vendors	30+	Microsoft
Interpreter	JRE	CLR
Dynamic Web Pages	JSP/JSF	ASP.NET
Middle-Tier Components	EJB	.NET Managed Components
Database access	JDBC, SQL/J	ADO.NET
SOAP, WSDL, UDDI	Yes	Yes
Implicit middleware (load-balancing, etc)	Yes	Yes

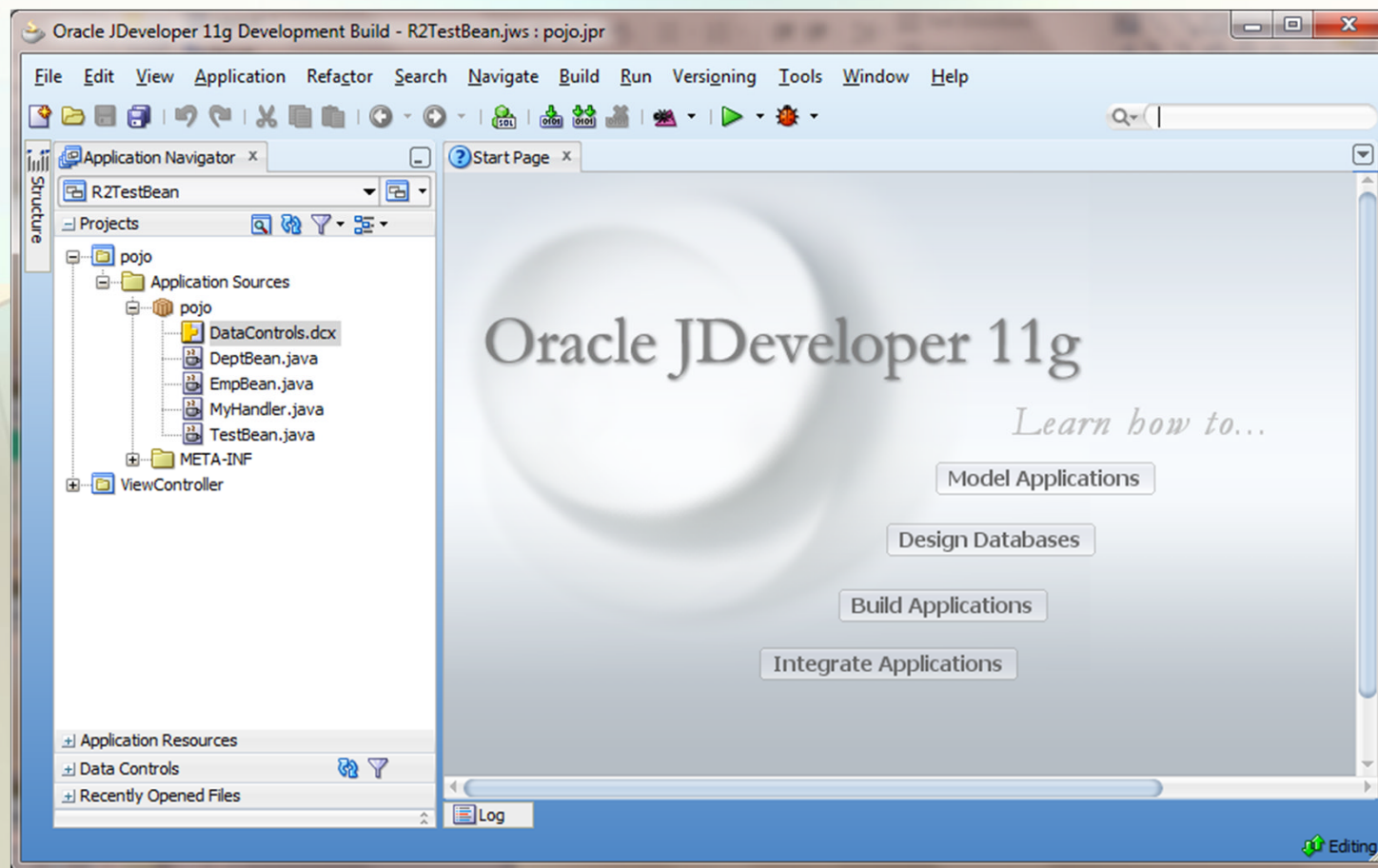


The J2EE and Microsoft.NET approach to Web Services



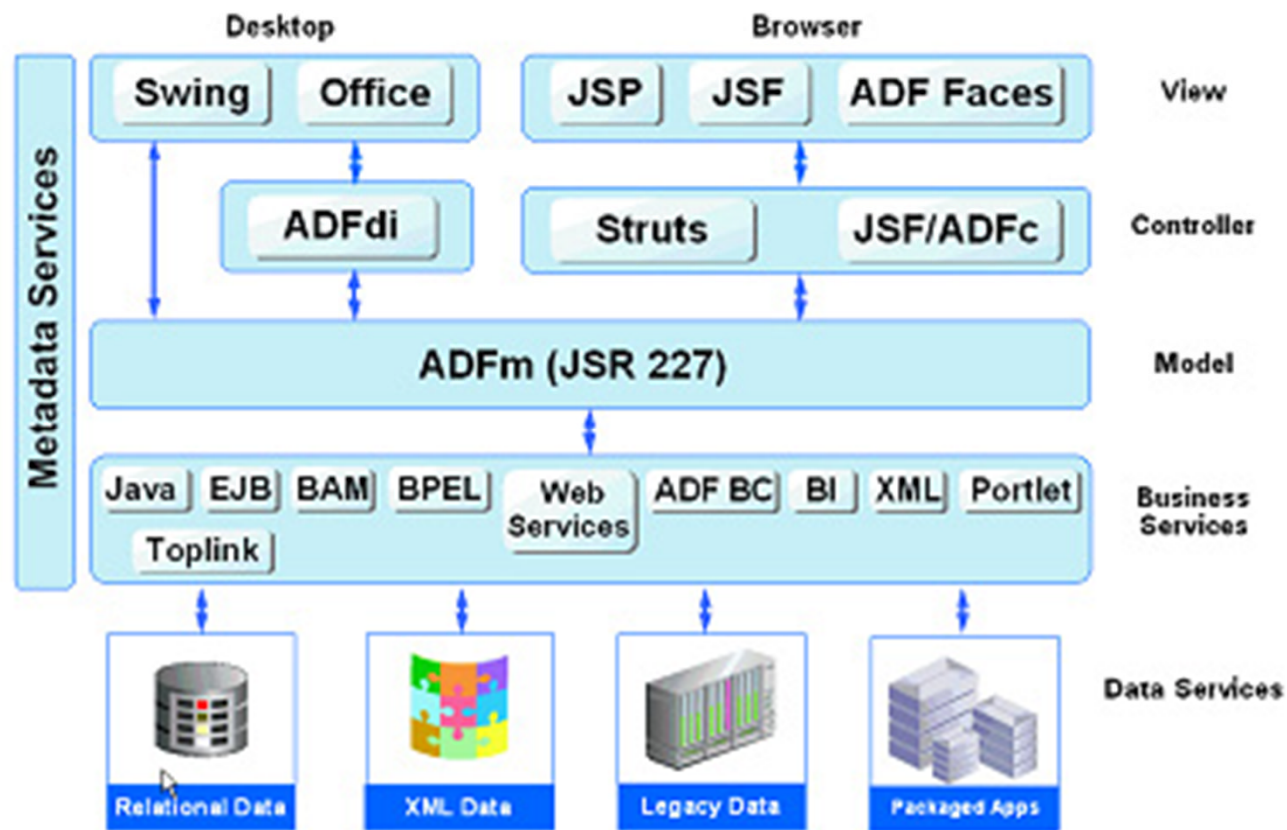


Java IDE: Oracle JDeveloper/ADF





Oracle JDeveloper/ADF

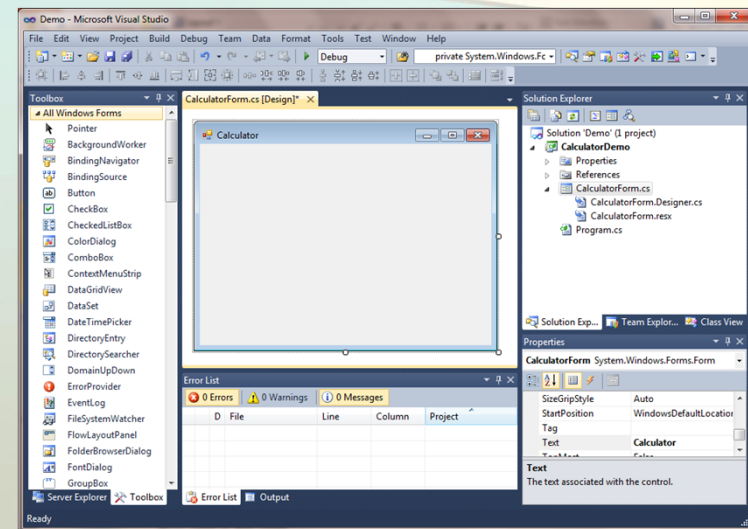




Microsoft Visual Studio Environment

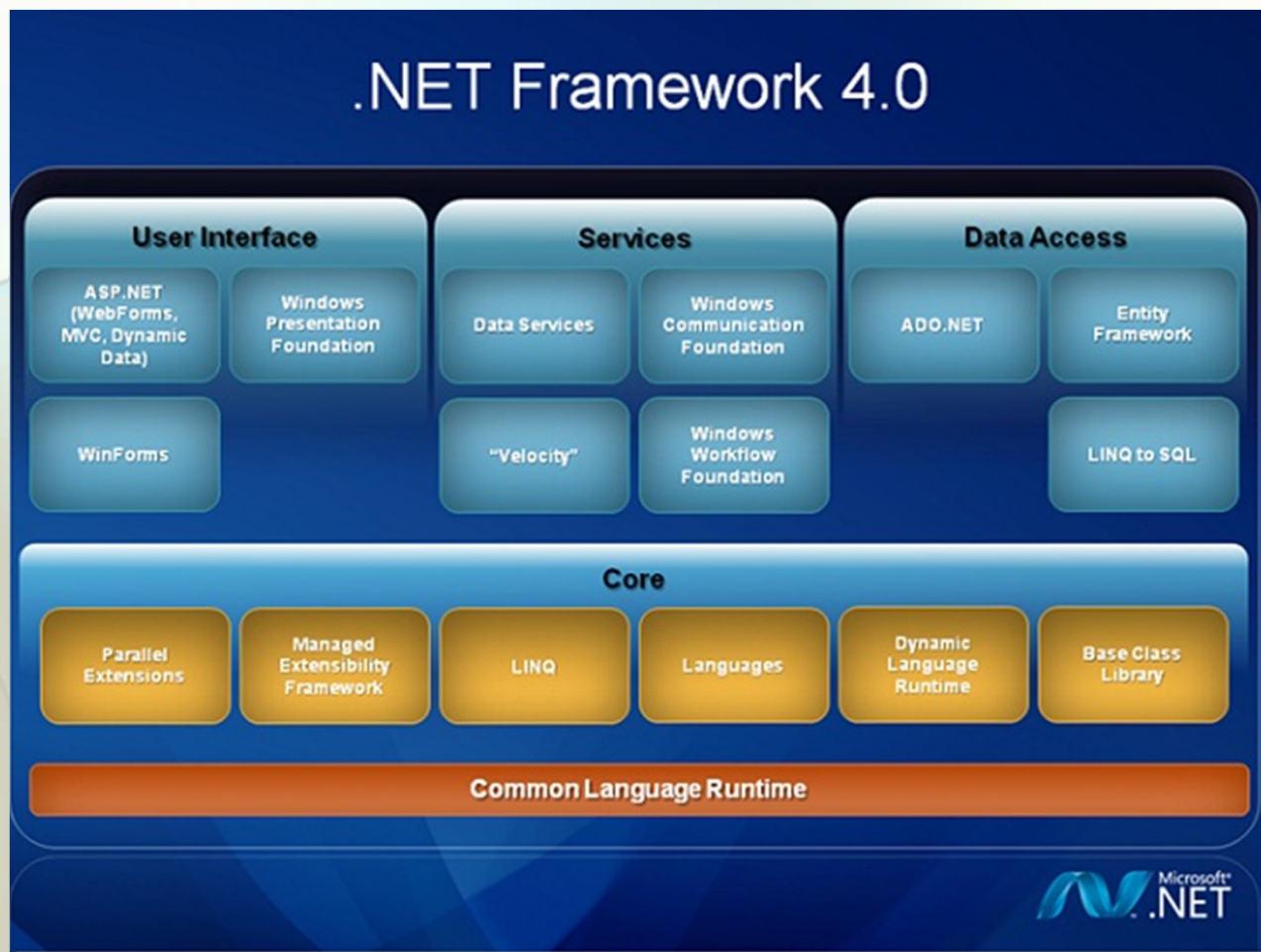
The Visual Studio environment is where you create and test your projects. In Visual Studio, it is called an

- Integrated Development Environment (IDE) consisting of various tools including:
- Form/WebForm Designer
- Editor for entering and modifying code
- Compiler
- Debugger
- Object Browser
- Help Facility





The Microsoft .NET Framework 4.0





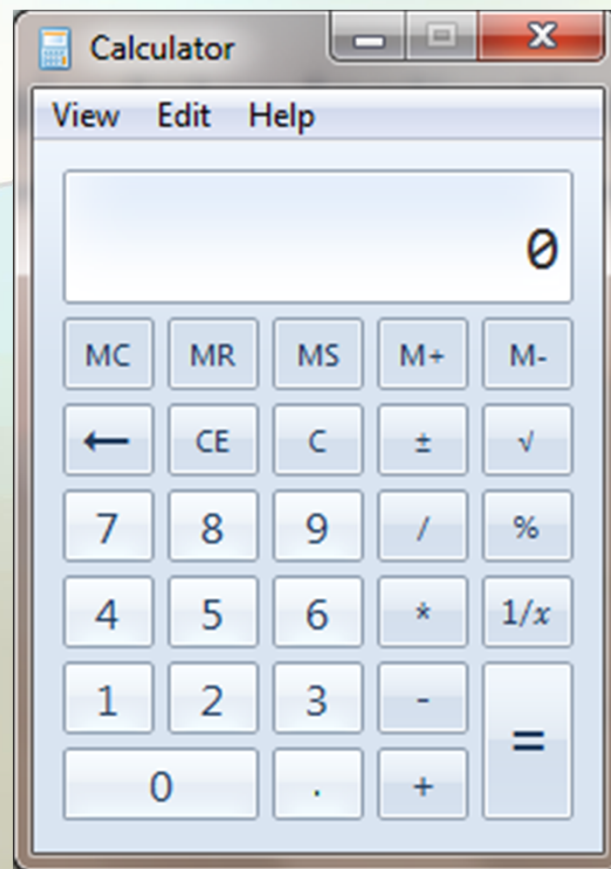
Overview of The Microsoft .NET Programming Languages

- Included in Visual Studio .NET
 - Visual Basic (VB.NET)
 - Visual C++
 - C# (C sharp)
 - J# (J sharp)
 - F# (F sharp)

.NET Framework 4 is the latest version



15 Minutes Application using The Microsoft .NET Framework





Q & A



Thank you!



Useful Links

- <http://msdn.microsoft.com/en-us/library/ms973223.aspx>
- <http://www.oracle.com/technetwork/developer-tools/jdev/overview/index.html>
- <http://developer.apple.com/technologies/mac/cocoa.html>
- <http://developer.android.com/sdk/index.html>
- <http://www.yolinux.com/TUTORIALS/LinuxTutorialSoftwareDevelopment.html>